



Addis Ababa University

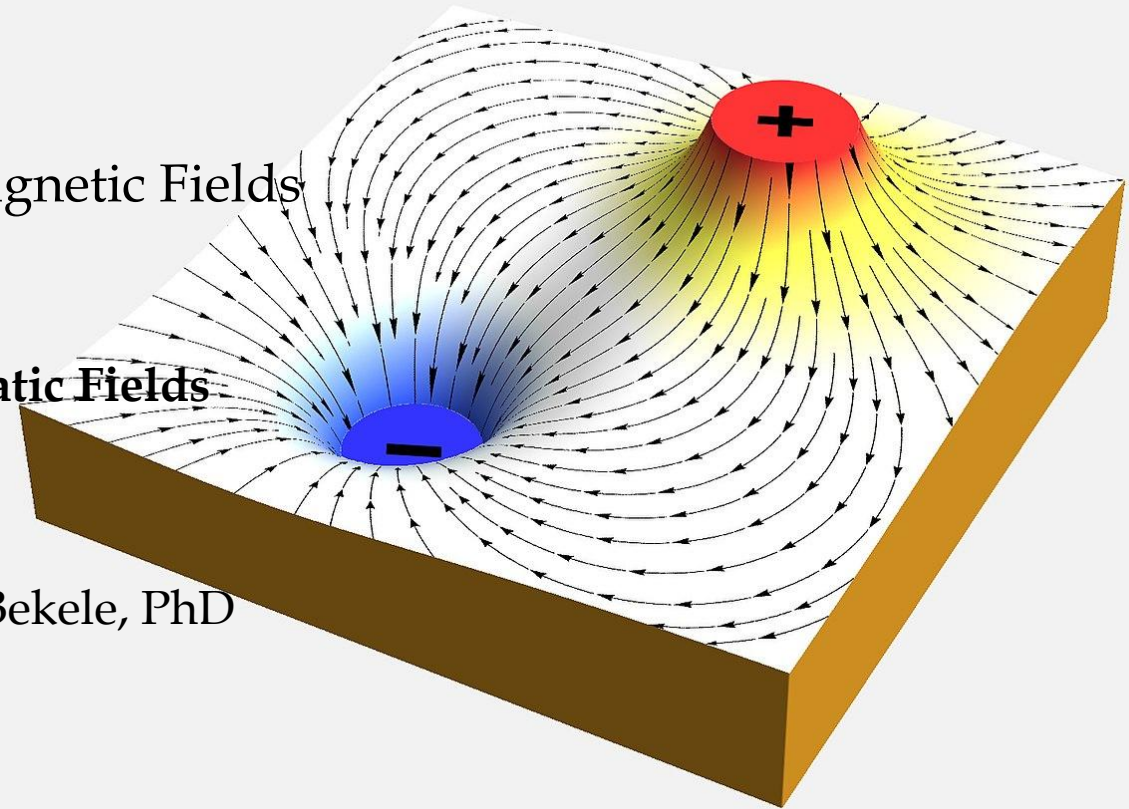
Collage of Technology and Built Environment (CTBE)

School of Electrical and Computer Engineering

ECEG 2102: Electromagnetic Fields

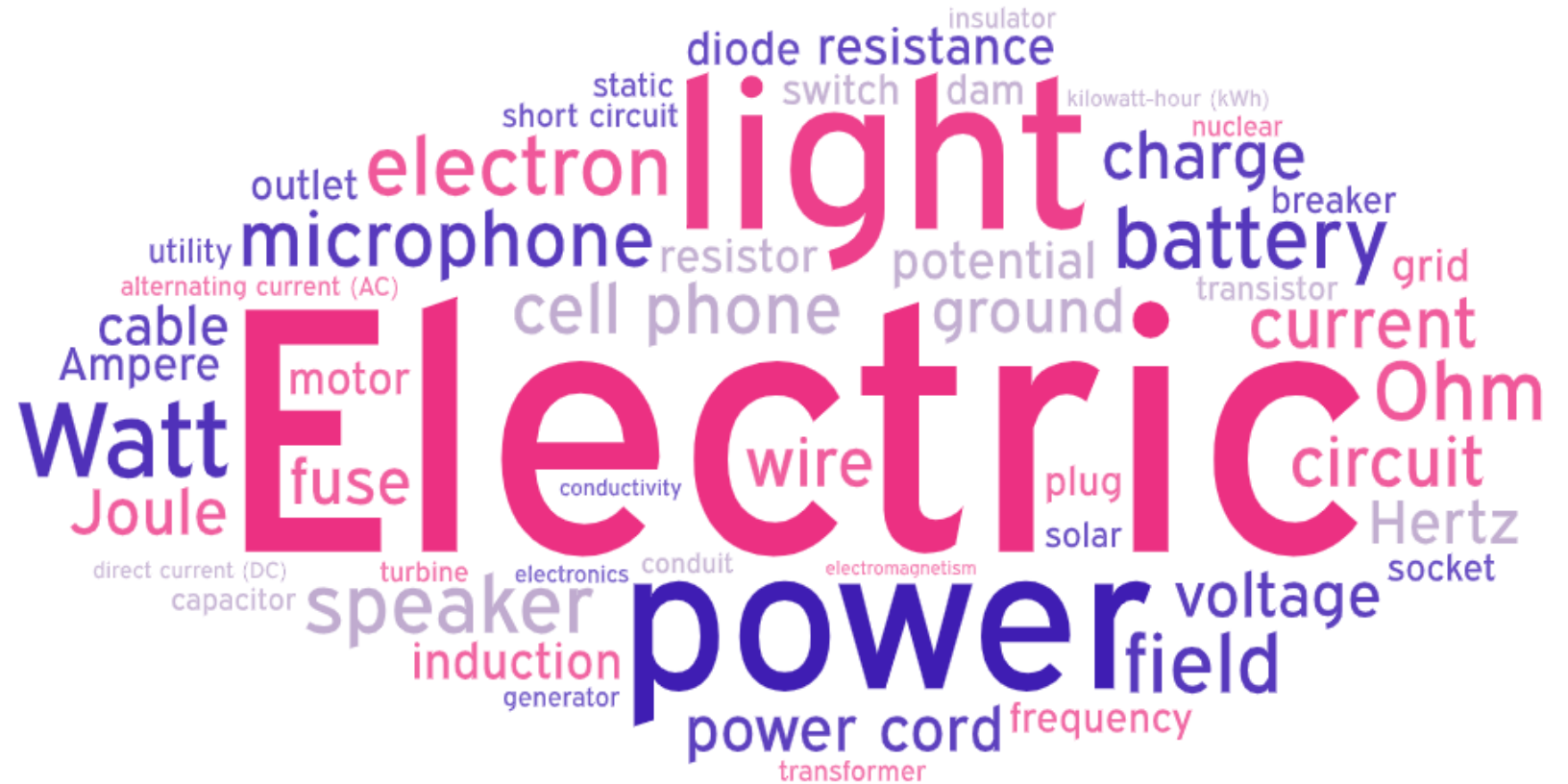
Module 3: Electrostatic Fields

2025 - Ephrem Teshale Bekele, PhD



What is Electricity?

The term **electrical** is in the name of the degree you seek to obtain when you graduate. But what is electricity?

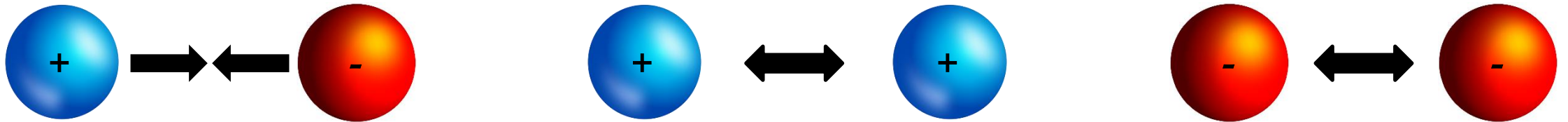


How would you explain it — in simple words — to a 10-year-old or your grandmother, without using technical jargon?



Electricity in Simple Words

- At the **most fundamental level**, electricity arises from **charged subatomic particles** — **protons** and **electrons**.
- Electrons carry negative charge and can **move between atoms**, while **protons remain** bound in the **atomic nucleus**.
- The **imbalance** or **movement** of these charges produces **observable electric phenomena**.
- In classical physics, we treat these charges as **point particles** whose interactions follow **Coulomb's Law**.
- Together, these microscopic interactions create **macroscopic effects** like **current flow**, **electric potential**, and **light emission**.
- Everything electrical — from **phones to power plants** — starts here.



Why This Matters to You

- Everything in electrical engineering — from power generation to signal transmission — is based on how **charges are distributed**, move, and **interact through electric fields**.
- The word **electric** traces back to the Greek “**ēlektron**,” meaning **amber** — where rubbing amber produced observable **static electricity**.
- Modern life depends on electricity:
 - The light that lets us work at night.
 - The internet that connects the world.
 - The machines and vehicles that power industries.
- In this course, we start with **electrostatics** — electricity at rest



What is Electrostatics?

- Electrostatics is the study of electric fields due to stationary charges.
- When charges do not move, they still exert forces on each other — forming the basis for electric fields.
- - Everyday phenomena of electrostatics include:
 - Static cling between clothes from a dryer.
 - Lightning caused by charge separation in clouds.
 - Balloons sticking to walls after being rubbed on hair.
- These examples show that even stationary charges create invisible regions of influence.





Coulomb's Law

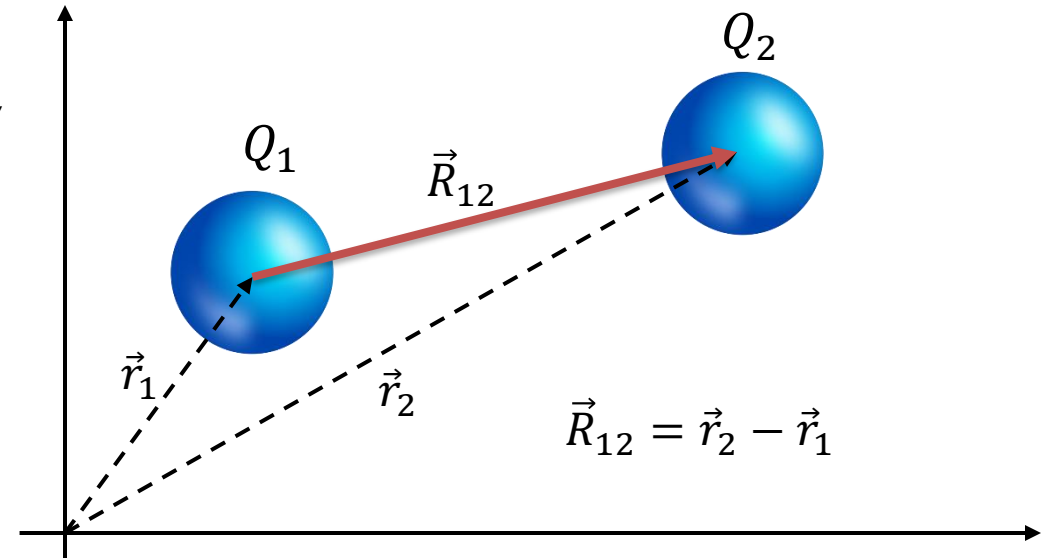
- Coulomb's law describes the **force between two stationary point charges**:

$$\vec{F}_{12} = \frac{Q_1 Q_2}{4\pi\epsilon_0 R_{12}^2} \vec{a}_{R_{12}}$$

- Force on Q_2 due to Q_1 , acts along the line from Q_1 to Q_2 .
- Like charges repel, unlike charges attract.**
- The strength of the force decreases with the **square of the distance**.
- $\epsilon_0 = 8.854 \times 10^{-12} \left[\frac{F}{m} \right]$ is the **permittivity of free space**, defining **how electric force behaves in vacuum**.
- Applications:
 - Determines **force between ions in a crystal**, **electrons in atoms**, or **charged conductors**.
 - Fundamental for computing electric fields from charge distributions.

- Since $\vec{R}_{12} = \vec{r}_2 - \vec{r}_1 \rightarrow R_{12} = |\vec{r}_2 - \vec{r}_1|$, $\vec{a}_{R_{12}} = \frac{\vec{R}_{12}}{|\vec{R}_{12}|} = \frac{(\vec{r}_2 - \vec{r}_1)}{|\vec{r}_2 - \vec{r}_1|}$:

$$\vec{F}_{12} = \frac{Q_1 Q_2}{4\pi\epsilon_0 |\vec{r}_2 - \vec{r}_1|^3} (\vec{r}_2 - \vec{r}_1)$$



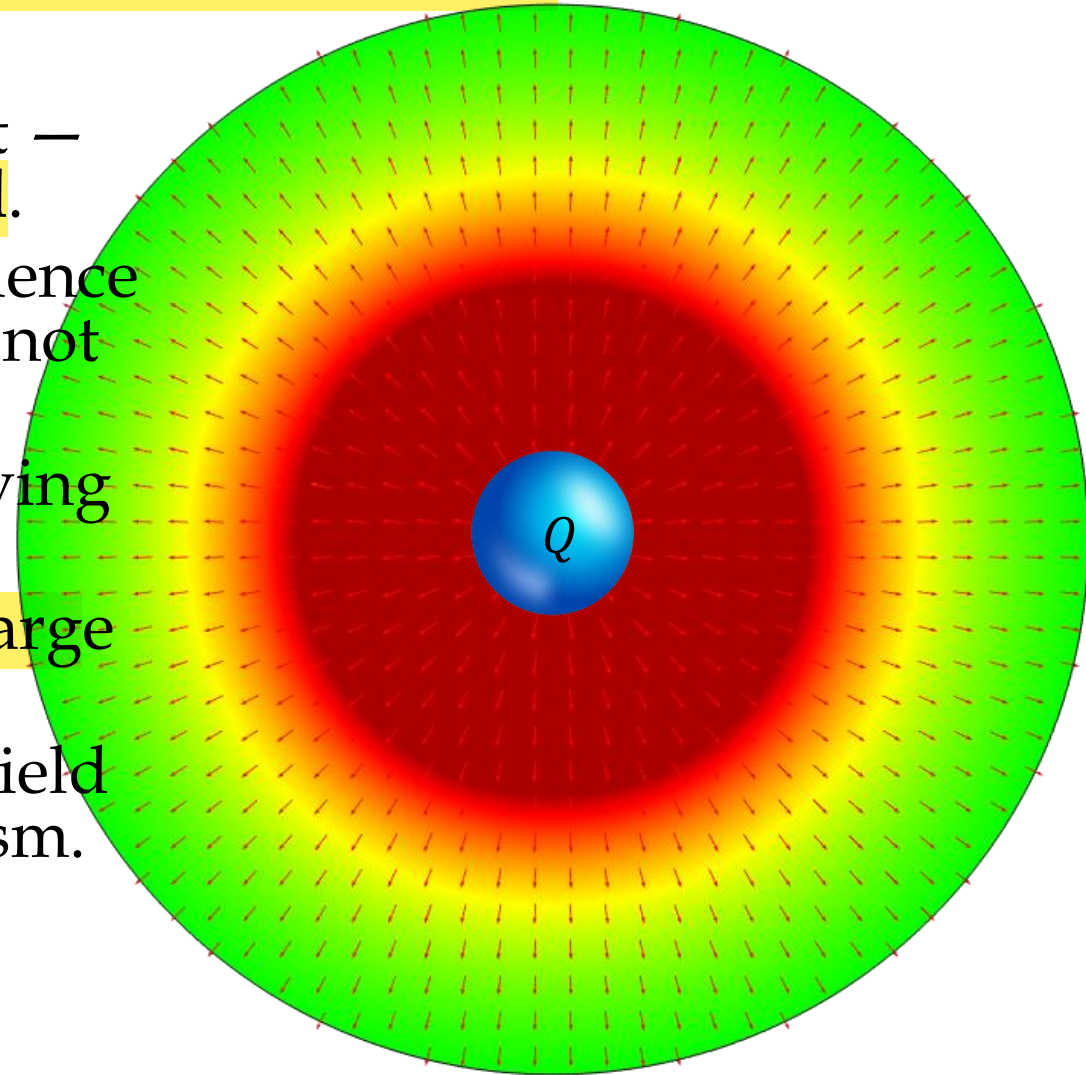
How Do Charges Interact?

- Imagine two charges, Q_1 and Q_2 .
- Q_2 feels a push or pull because of Q_1 — but how does it “know” Q_1 exists?
- If Q_2 could talk, it might say:
 - “Hey Q_1 , I can feel your push — stop it!”
- Yet, they never touch each other directly.
- So how does one charge affect another **across empty space**?



From Direct Forces to Fields

- Every charge modifies the space around it — this modification is called an **electric field**.
- Other charges placed in that region experience a force because they respond to this field, not directly to the distant charge.
- Thus, the field acts as the **messenger** carrying the influence of one charge to another.
- The field exists everywhere, even if no charge is there to perceive it.
- This idea — that charges act through the field they create — is central to electromagnetism.



Electric Field Intensity ($\vec{E}$)

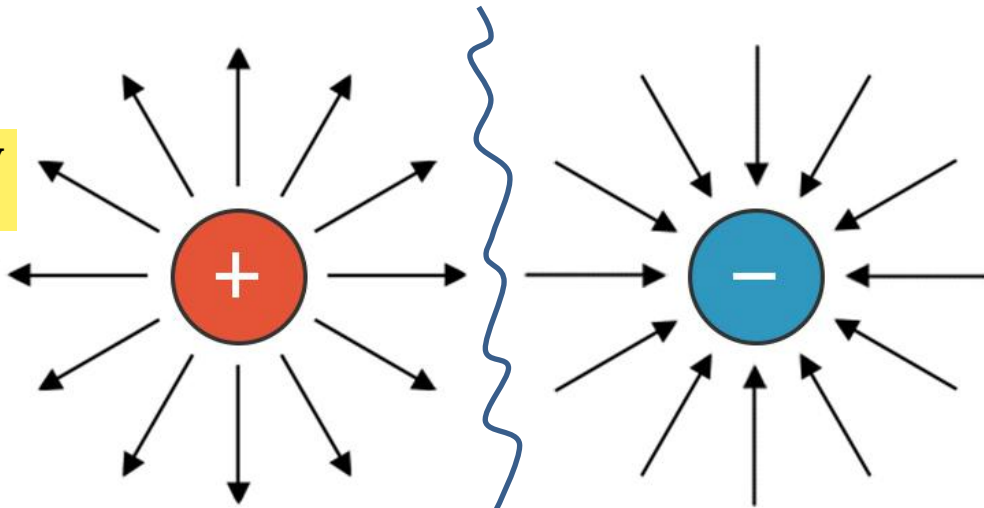
- The **electric field intensity**, ($\vec{E}$), is defined as the **force per unit positive charge**:

$$\vec{E} = \frac{\vec{F}}{q}.$$

- Units: **N/C or V/m**.
- The field describes how a charge influences the space around it.
- The electric field intensity at point $\vec{r}$ due to a point charge Q located at $\vec{r}'$ is

$$\vec{E} = \frac{Q}{4\pi\epsilon_0|\vec{r} - \vec{r}'|^3}(\vec{r} - \vec{r}')$$

- Field lines, hypothetical lines showing presence of $\vec{E}$, point **away** from positive charges and **toward** negative charges.
- The **density of lines** represents the field's strength – closer lines mean stronger fields.
- The concept of **field lines** was introduced by **Michael Faraday** in the 1830s.
- He visualized electric and magnetic forces as lines of influence filling space, providing the foundation for the modern field theory developed later by Maxwell.

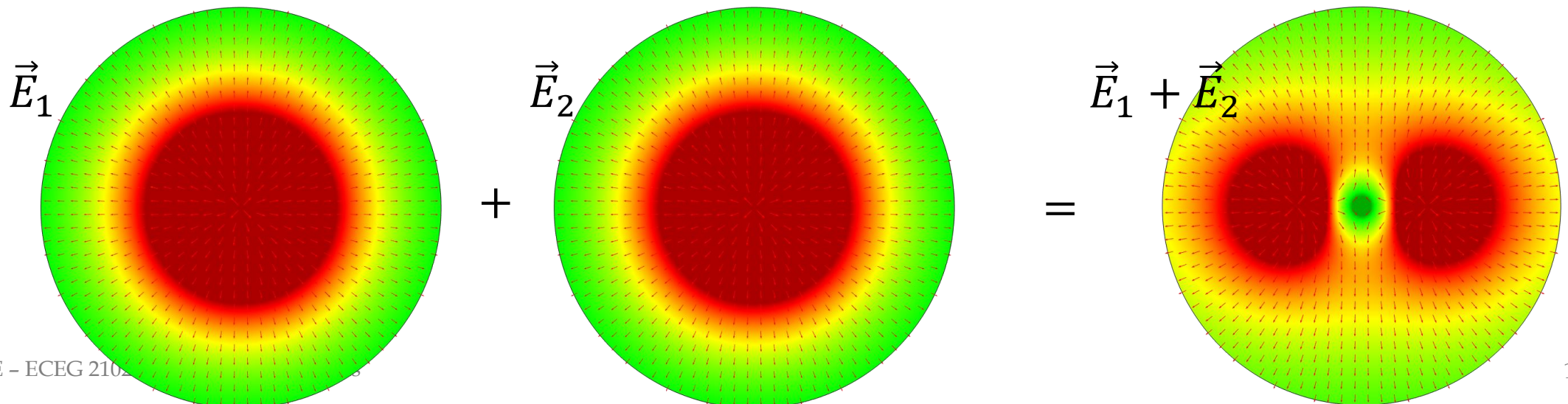


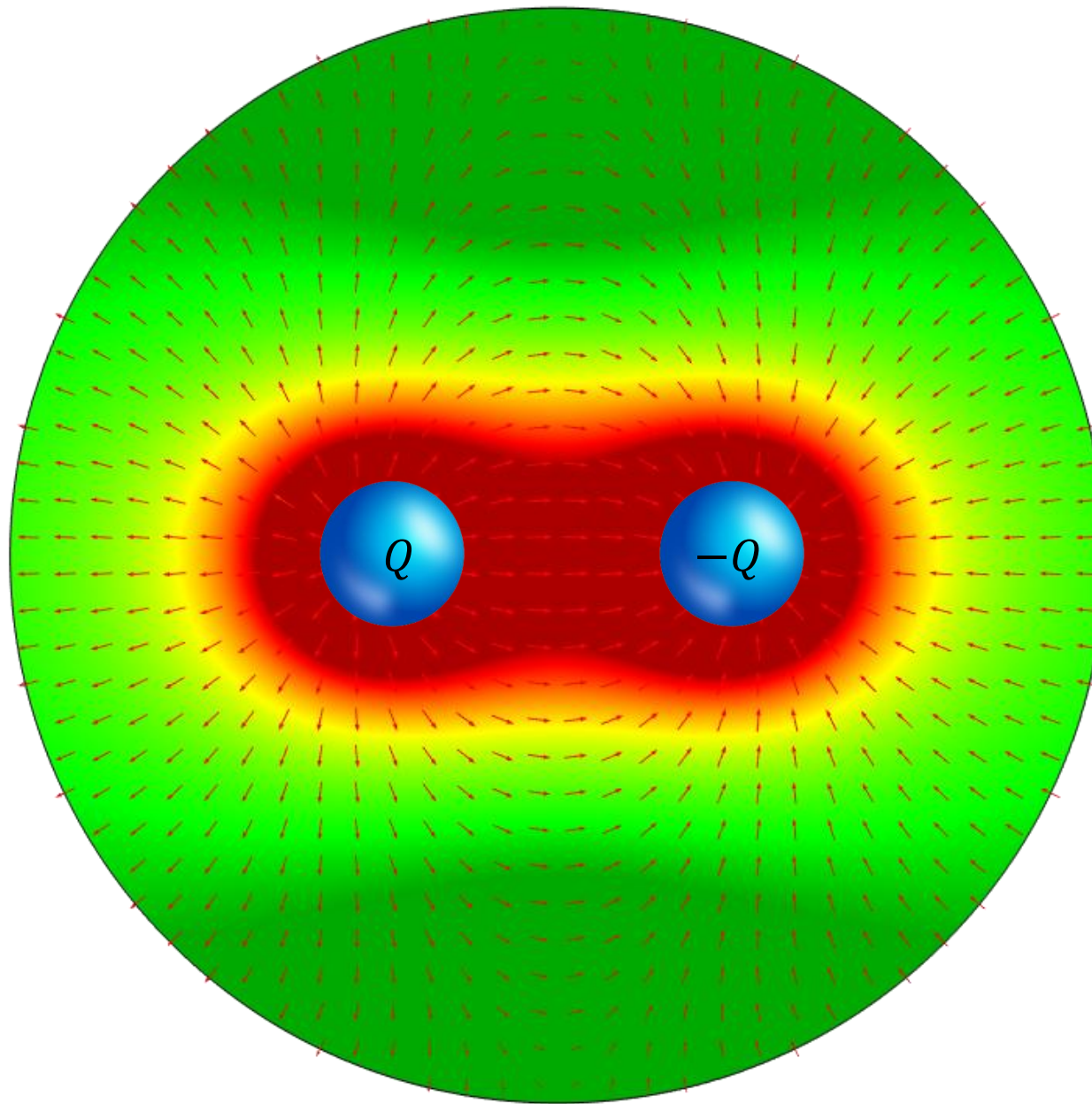
Superposition Principle

- When multiple charges exist, each creates its own field.
- The total electric field is the **vector sum** of all individual fields:

$$\vec{E} = \sum_i \vec{E}_i = \frac{1}{4\pi\epsilon_0} \sum_i \frac{Q_i}{|\vec{r} - \vec{r}'_i|^3} (\vec{r} - \vec{r}'_i)$$

- This principle simplifies analysis of complex charge arrangements.
- The direction and magnitude of the resultant field depend on relative positions and magnitudes of all charges.





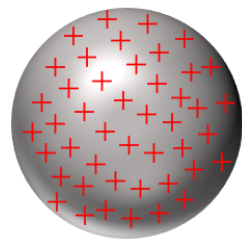
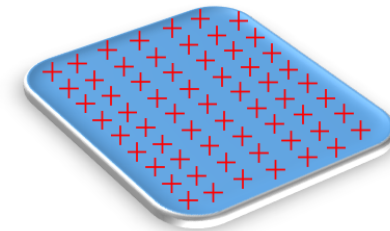
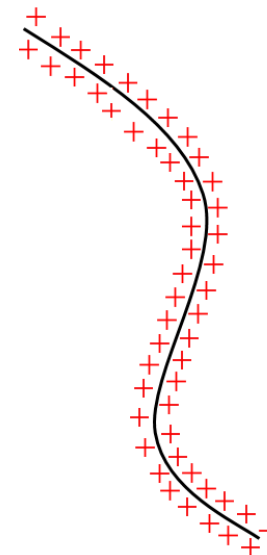
Continuous Charge Distributions

- Realistic systems contain many charges, it's more practical to treat them as **continuous charge densities**.
- Line charge density: ρ_l (C/m) $\rightarrow$ distributed along a line.
- Surface charge density: ρ_s (C/m²) $\rightarrow$ spread over an area.
- Volume charge density: ρ_v (C/m³) $\rightarrow$ fills a region of space.
- The net electric field is found using integration over the region:

$$\vec{E} = \int_{l'} \frac{\rho_l}{4\pi\epsilon_0 R^2} \vec{a}_R dl' = \int_{l'} \frac{\rho_l}{4\pi\epsilon_0 |\vec{r} - \vec{r}'|^3} (\vec{r} - \vec{r}') dl'$$

$$\vec{E} = \iint_{S'} \frac{\rho_s}{4\pi\epsilon_0 R^2} \vec{a}_R dS = \iint_S \frac{\rho_s}{4\pi\epsilon_0 |\vec{r} - \vec{r}'|^3} (\vec{r} - \vec{r}') dS'$$

$$\vec{E} = \iiint_{v'} \frac{\rho_v}{4\pi\epsilon_0 R^2} \vec{a}_R dv' = \iiint_{v'} \frac{\rho_v}{4\pi\epsilon_0 |\vec{r} - \vec{r}'|^3} (\vec{r} - \vec{r}') dv'$$



Classroom Exercise (1/2)

1. Derive the following commonly used electric field intensity formulas:

a) A finite length line charge with uniform charge density ρ_l

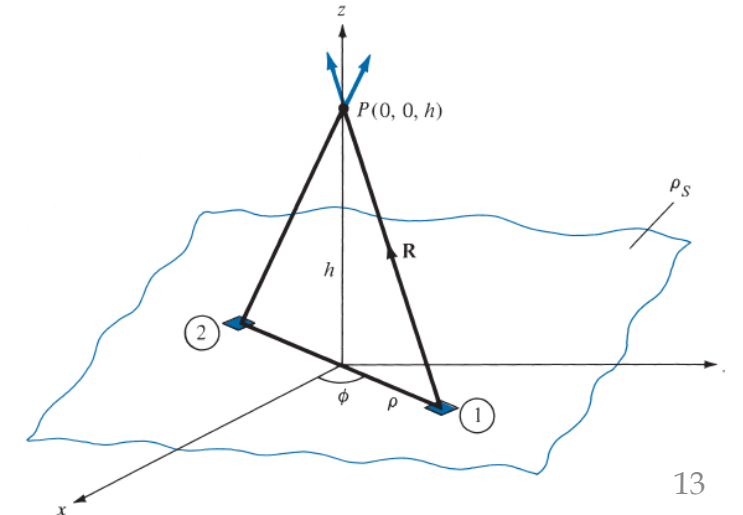
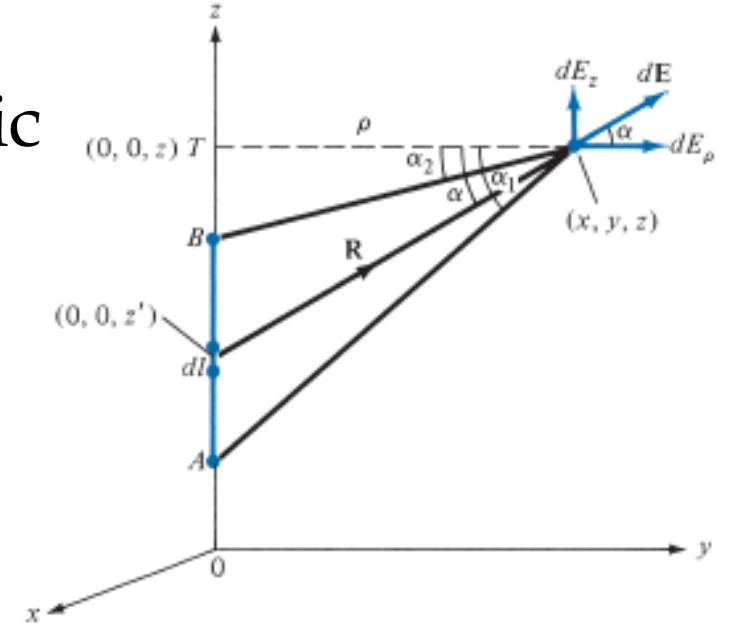
$$\vec{E} = \frac{\rho_l}{4\pi\epsilon_0\rho} [-(\sin \alpha_2 - \sin \alpha_1)\vec{a}_\rho + (\cos \alpha_2 - \cos \alpha_1)\vec{a}_z]$$

b) Infinite line charge

$$\vec{E} = \frac{\rho_l}{2\pi\epsilon_0\rho} \vec{a}_\rho$$

c) Infinite surface charge with uniform charge density ρ_s

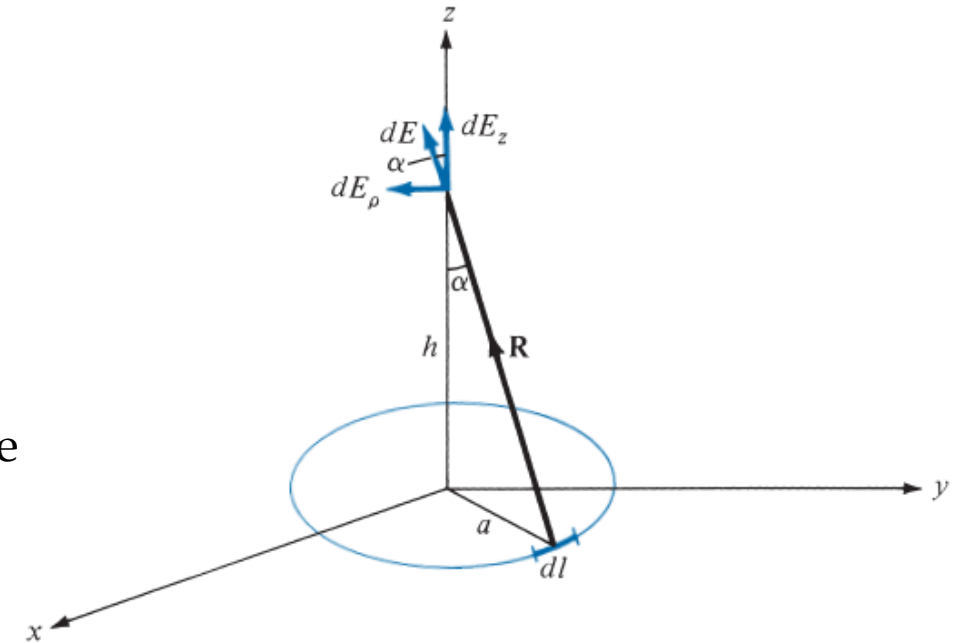
$$\vec{E} = \frac{\rho_s}{2\epsilon_0} \vec{a}_n$$



Classroom Exercise (2/2)

2. A circular ring of radius a carries a uniform charge $\rho_l \left[\frac{C}{m} \right]$ and is placed on the xy-plane with axis the same as the z-axis.

- Show that $\vec{E}(0,0,h) = \frac{\rho_l a h}{2\epsilon_0 (h^2 + a^2)^{3/2}} \vec{a}_z$
- What value of h gives the maximum value of $\vec{E}$ (Ans. $h = \pm \frac{h}{\sqrt{a}}$)
- If the total charge on the ring is Q , find $\vec{E}$ as $a \rightarrow 0$. Compare the result with that of a point charge and state your observations.
(Ans. $\vec{E}(0,0,h) = \frac{Q}{4\pi\epsilon_0 h^2} \vec{a}_z$)



3. Planes $x = 2$ and $y = 3$ respectively carry charges $10 \frac{nC}{m^2}$ and $15 \frac{nC}{m^2}$. If the line $x = 0; z = 2$ carries charge $10 \frac{nC}{m}$, calculate $\vec{E}$ at $(1,1,1)$ due to the three charge distributions.

Classroom Exercise – Solution (1/2)

Finite Length Line Charge

- Consider a line charge with uniform charge density ρ_l extending from A to B along the z -axis as shown in Figure.
- The charge element dQ associated with element $dl' = dz'$ of the line is:

$$dQ = \rho_l dl' = \rho_l dz'$$

- The electric field intensity $\vec{E}$ at an arbitrary point P $\vec{r} = (x, y, z)$ can be found by using the line integral equation: $\vec{E} = \int_{l'} \frac{\rho_l}{4\pi\epsilon_0 R^2} \vec{a}_R dl' = \int_{l'} \frac{\rho_l}{4\pi\epsilon_0 |\vec{r} - \vec{r}'|^3} (\vec{r} - \vec{r}') dl'$ where the source point is represented by primes $\vec{r}' = (x', y', z')$. Thus:

$$dl' = dz'$$

$$\vec{R} = \vec{r} - \vec{r}' = (x, y, z) - (0, 0, z') = x\vec{a}_x + y\vec{a}_y + (z - z')\vec{a}_z$$

- Considering cylindrical symmetry of the problem, let us rewrite the quantities in cylindrical coordinates and quantities shown in the figure:

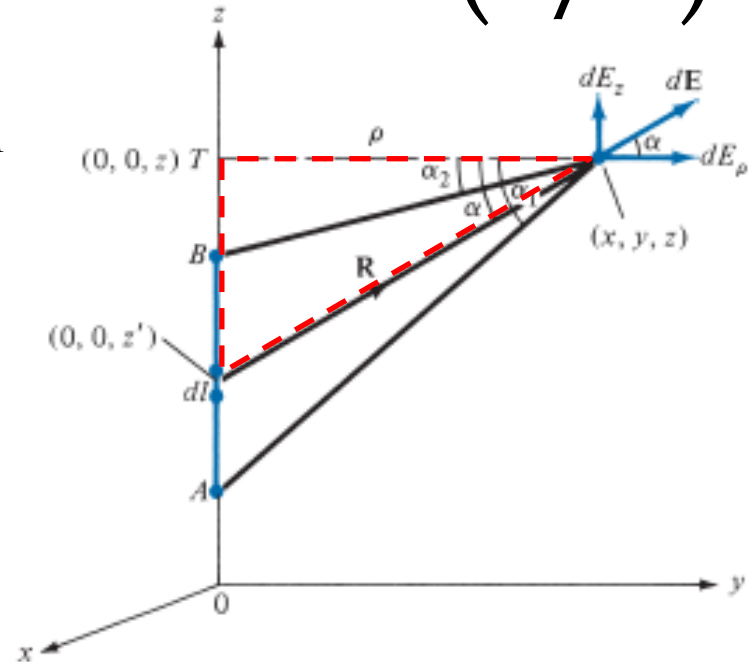
$$\vec{r} - \vec{r}' = \vec{R} = \rho\vec{a}_\rho + (z - z')\vec{a}_z$$

Considering the right angle triangle shown in red:

$$z - z' = \rho \tan \alpha \Rightarrow z' = z - \rho \tan \alpha \Rightarrow dz' = \frac{-\rho}{\cos^2 \alpha} d\alpha$$

$$\vec{r} - \vec{r}' = \vec{R} = \rho\vec{a}_\rho + \rho \tan \alpha \vec{a}_z$$

$$|\vec{r} - \vec{r}'| = R = \frac{\rho}{\cos \alpha}$$



Hence:

$$\begin{aligned} \vec{E} &= \int_{l'} \frac{\rho_l}{4\pi\epsilon_0 |\vec{r} - \vec{r}'|^3} (\vec{r} - \vec{r}') dl' \\ &= \int_{\alpha_1}^{\alpha_2} \frac{\rho_l}{4\pi\epsilon_0 \left(\frac{\rho}{\cos \alpha}\right)^3} (\rho\vec{a}_\rho + \rho \tan \alpha \vec{a}_z) \left(\frac{-\rho}{\cos^2 \alpha} d\alpha\right) \\ &= \frac{-\rho_l}{4\pi\epsilon_0 \rho} \int_{\alpha_1}^{\alpha_2} (\cos \alpha \vec{a}_\rho + \sin \alpha \vec{a}_z) d\alpha = \frac{-\rho_l}{4\pi\epsilon_0 \rho} (\sin \alpha \vec{a}_\rho - \cos \alpha \vec{a}_z) \Big|_{\alpha_1}^{\alpha_2} \\ &= \frac{-\rho_l}{4\pi\epsilon_0 \rho} [(\sin \alpha_2 \vec{a}_\rho - \cos \alpha_2 \vec{a}_z) - (\sin \alpha_1 \vec{a}_\rho - \cos \alpha_1 \vec{a}_z)] \\ &= \frac{-\rho_l}{4\pi\epsilon_0 \rho} [\sin \alpha_2 \vec{a}_\rho - \cos \alpha_2 \vec{a}_z - \sin \alpha_1 \vec{a}_\rho + \cos \alpha_1 \vec{a}_z] \\ &= \frac{\rho_l}{4\pi\epsilon_0 \rho} [-(\sin \alpha_2 - \sin \alpha_1) \vec{a}_\rho + (\cos \alpha_2 - \cos \alpha_1) \vec{a}_z] \end{aligned}$$

Classroom Exercise – Solution (2/2)

Infinite Length Line Charge

- When the line becomes infinitely large:

$$\alpha_1 \rightarrow \frac{\pi}{2} \text{ and } \alpha_2 \rightarrow -\frac{\pi}{2}$$

$$\vec{E} = \frac{\rho_l}{2\pi\epsilon_0\rho}$$

Infinite Surface Charge

- Consider an infinite sheet of charge in the xy -plane with uniform charge density ρ_s .
- The charge associated with an elemental area dS' is $dQ = \rho_s dS'$
- The contribution to the $\vec{E}$ field at point $P \vec{r} = (0,0,h)$ by the charge dQ on the elemental surface 1 is:

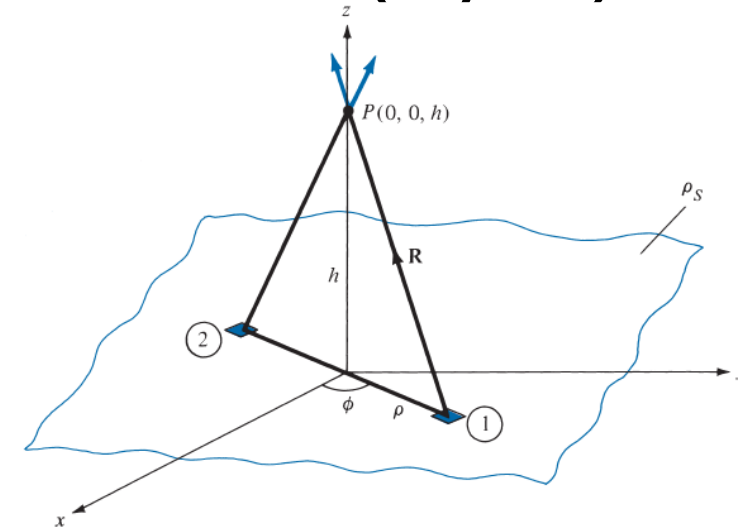
$$d\vec{E} = \frac{dQ}{4\pi\epsilon_0} \frac{\vec{R}}{|\vec{R}|^3}$$

- where, $\vec{R} = \rho'(-\vec{a}_\rho) + h\vec{a}_z$, $dS' = \rho' d\phi' d\rho'$.

$$d\vec{E} = \frac{\rho_s \rho' d\phi' d\rho'}{4\pi\epsilon_0} \frac{(-\rho' \vec{a}_\rho + h \vec{a}_z)}{(\rho'^2 + h^2)^{\frac{3}{2}}}$$

- Owing to the symmetry of the charge distribution, for every element 1, there is a corresponding element 2 whose contribution along $\vec{a}_\rho$ cancels that of element 1. Thus the contributions to E_ρ add up to zero so that $\vec{E}$ has only z -component.

$$\vec{E} = \int_{S'} dE_z \vec{a}_z = \int_{\phi'=0}^{2\pi} \int_{\rho'=0}^{\infty} \frac{\rho_s \rho' d\phi' d\rho'}{4\pi\epsilon_0} \frac{h \vec{a}_z}{(\rho'^2 + h^2)^{\frac{3}{2}}} = \frac{\rho_s h \vec{a}_z}{4\pi\epsilon_0} \int_{\phi'=0}^{2\pi} \int_{\rho'=0}^{\infty} \frac{\rho'}{(\rho'^2 + h^2)^{\frac{3}{2}}} d\rho' d\phi'$$



- Substitute: $u = \rho'^2 + h^2$, $du = 2\rho' d\rho'$

$$\int_{\rho'=0}^{\infty} \frac{\rho'}{(\rho'^2 + h^2)^{\frac{3}{2}}} d\rho' = \int_{u=h^2}^{\infty} \frac{1}{2} u^{-\frac{3}{2}} du = -u^{-\frac{1}{2}} \Big|_{h^2}^{\infty} = \frac{1}{h}$$
- Hence:

$$\vec{E} = \frac{\rho_s h \vec{a}_z}{4\pi\epsilon_0} \int_{\phi'=0}^{2\pi} \int_{\rho'=0}^{\infty} \frac{\rho'}{(\rho'^2 + h^2)^{\frac{3}{2}}} d\rho' d\phi' = \frac{\rho_s h \vec{a}_z}{4\pi\epsilon_0} \int_{\phi'=0}^{2\pi} \frac{1}{h} d\phi'$$

$$= \frac{\rho_s h \vec{a}_z}{4\pi\epsilon_0} \frac{2\pi}{h} = \frac{\rho_s}{2\epsilon_0} \vec{a}_z$$
- Note that the electric field is constant, **independent of the distance** from the surface.

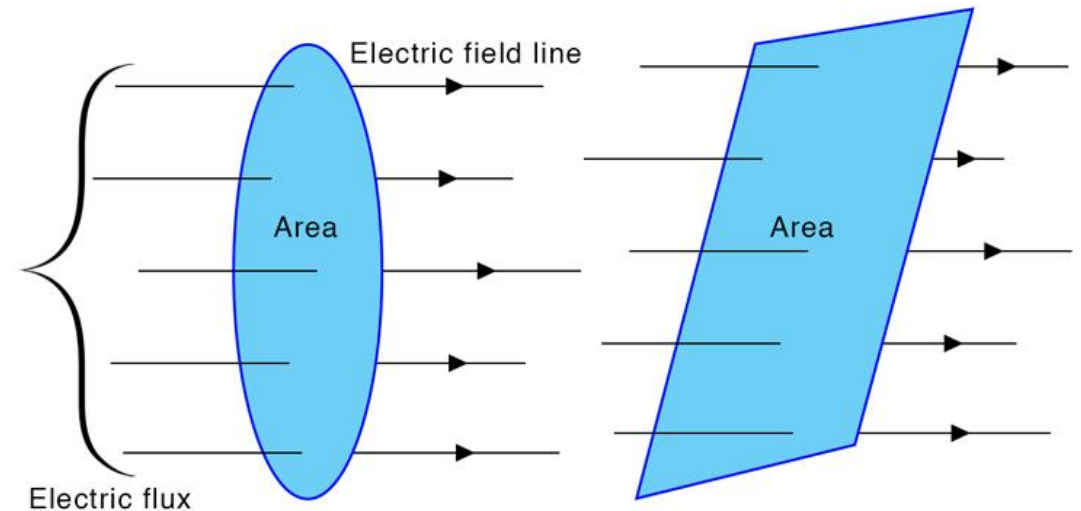
Electric Flux Density ($\vec{D}$)

- The electric flux density ($\vec{D}$) is defined as ($\vec{D} = \epsilon \vec{E}$).
- Unit: C/m².
- Represents the amount of “electric field flow” through a surface.
- For point charges:

$$\vec{D} = \epsilon \vec{E} = \frac{Q}{4\pi |\vec{r} - \vec{r}'|^3} (\vec{r} - \vec{r}')$$

- In dielectric materials, $\vec{D}$ links the electric field to charge behavior within the medium.
- The electric flux through a surface S is defined as:

$$\Psi = \int_S \vec{D} \cdot d\vec{S}$$



Classroom Exercise

The line $y = 1; z = -3$ carries charge 30 nC/m while the plane $x = 1$ carries charge 20 nC/m^2 . Find $\vec{D}$ at the origin.
(Ans. $-10.01\vec{a}_x - 0.478\vec{a}_y + 1.434\vec{a}_z \text{ nC/m}^2$)



Gauss's Law

- The total electric flux through any closed surface equals the charge enclosed within it.

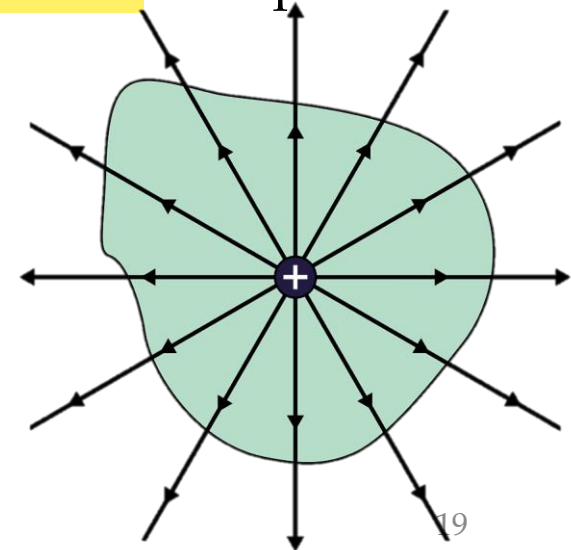
$$\Psi = \oint \vec{D} \cdot d\vec{S} = Q_{enc}$$

- Since $Q_{enc} = \int_V \rho_v dV \rightarrow \oint \vec{D} \cdot d\vec{S} = \int_V \rho_v dV$, and from Divergence theorem, $\oint \vec{D} \cdot d\vec{S} = \int_V \nabla \cdot \vec{D} dV$, we can establish the Differential Form of Gauss's law:

$$\nabla \cdot \vec{D} = \rho_v$$

which states that at any given point, the divergence of the electric flux lines is equal to the volume charge density at that point.

- Gauss's law serves as a powerful alternative to Coulomb's law for symmetric problems.
- The direction and uniformity of field lines depend on geometry (spherical, cylindrical, planar).
- Gauss's Law is one of the components of Maxwell's Equations.



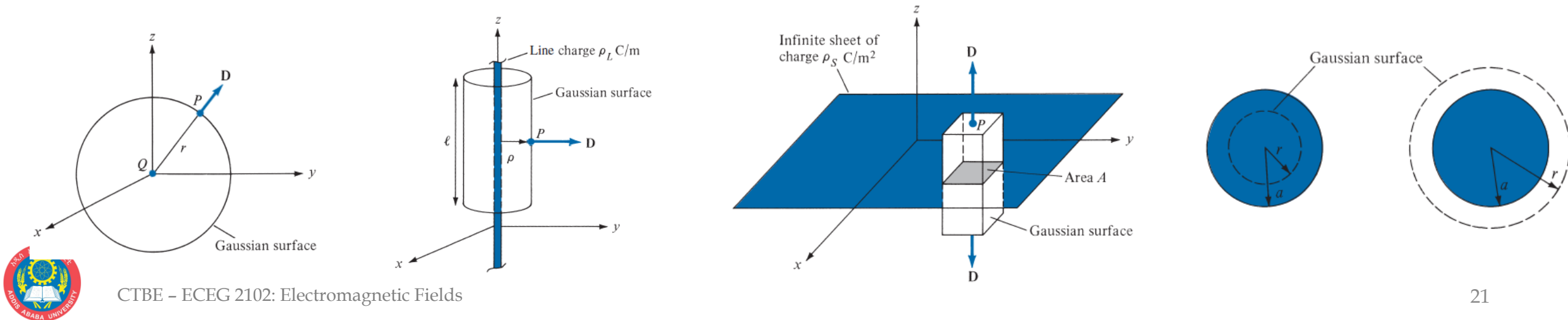
Gauss's Law – Applications

- Simplifies field calculations for symmetrical charge distributions:
 - Point charge → spherical symmetry.
 - Infinite line charge → cylindrical symmetry.
 - Infinite plane → planar symmetry.
- Steps:
 1. Identify whether there is symmetry exists.
 2. Construct a mathematical closed surface (Gaussian surface).
 - The surface is chosen such that $\vec{D}$ is normal or tangential to the Gaussian surface.
 - When $\vec{D}$ is normal to the surface, $\vec{D} \cdot d\vec{S} = DdS$.
 - When $\vec{D}$ is tangential to the surface, $\vec{D} \cdot d\vec{S} = 0$.
 3. Apply $\oint \vec{D} \cdot d\vec{S} = Q_{enc}$.
 4. Solve for $\vec{D}$ or $\vec{E}$.



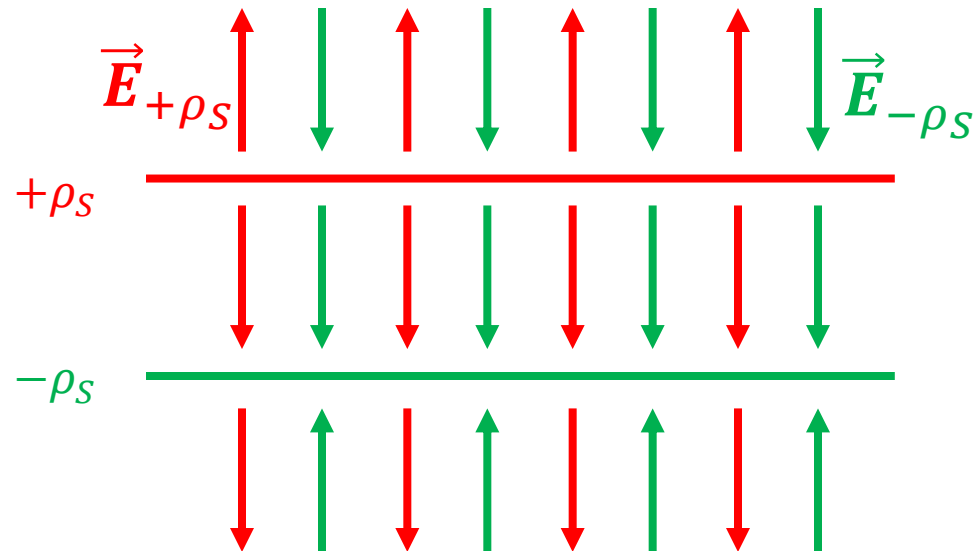
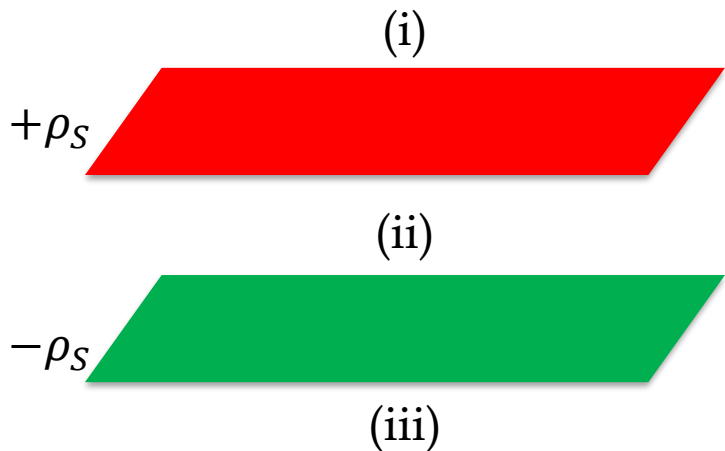
Gauss's Law – Examples (1/3)

- Using Gauss's law, derive $\vec{D}$ and $\vec{E}$ for:
 - point charge
 - infinite line charge
 - infinite sheet of charge
- Consider a sphere of radius a with uniform charge $\rho_v \left[\frac{C}{m} \right]$. Find $\vec{D}$ and $\vec{E}$ for $r < a$ and $r > a$. For $r > a$, express the result in terms of total charge Q and explain your observation.



Gauss's Law – Examples (2/3)

- Two infinite parallel planes carry equal but opposite uniform charge densities $\pm\rho_S$.
 - Find the field in each of the following three regions.
 - Compare the results with formula for electric field inside a parallel plate capacitor and state your observations.



Gauss's Law – Examples (3/3)

- A thick spherical shell carries charge density

$$\rho = \frac{k}{r^2} \quad a \leq r \leq b$$

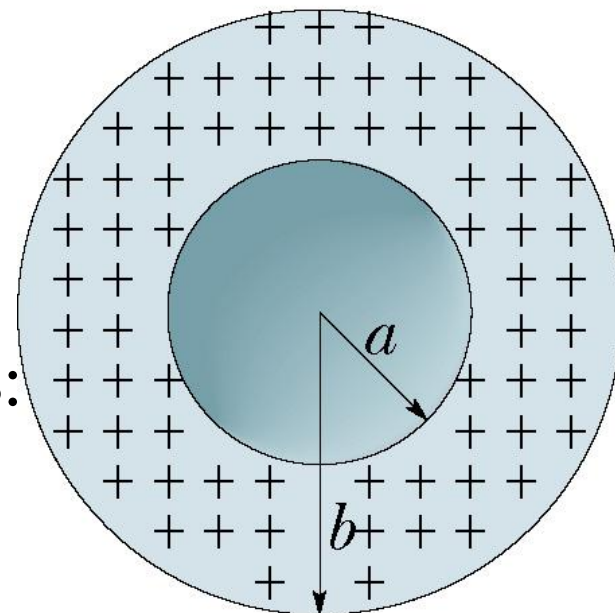
Find the electric field in the following three regions:

(i) $r < a$,

(ii) $a < r < b$,

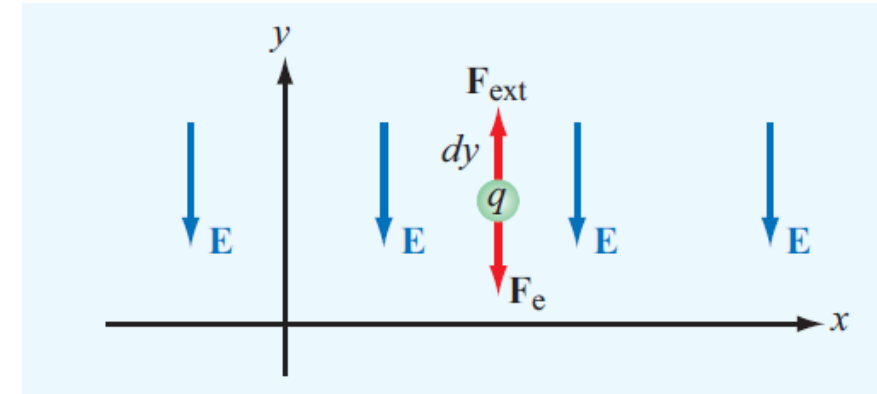
(iii) $r > b$. In this case express the electric field in terms of the total charge in the spherical shell and discuss your observation.

Plot $|\vec{E}|$ as a function of r , for the case $b = 2a$.



Electric Potential (1/2)

- Electric potential represents the work done in moving a unit charge between two points in an Electric field.
- Referring to the figure, the presence of the field $\vec{E}$ exerts a force $\vec{F}_e = q\vec{E}$ on the charge.
- To move the charge against the force $\vec{F}_e$, we need to provide an external force $\vec{F}_{ext}$ to counteract $\vec{F}_e$, which requires the expenditure of energy.
- To move q without acceleration, the net force acting on the charge must be zero: $\vec{F}_{ext} + \vec{F}_e = 0$, or $\vec{F}_{ext} = -\vec{F}_e = -q\vec{E}$.



- The work done in moving an object a differential distance $d\vec{l}$ while exerting a force $\vec{F}_{ext}$ is
$$dW = \vec{F}_{ext} \cdot d\vec{l} = -q\vec{E} \cdot d\vec{l} [J].$$
- The differential electric potential energy dW per unit charge is called the differential electric potential (or differential voltage) dV :

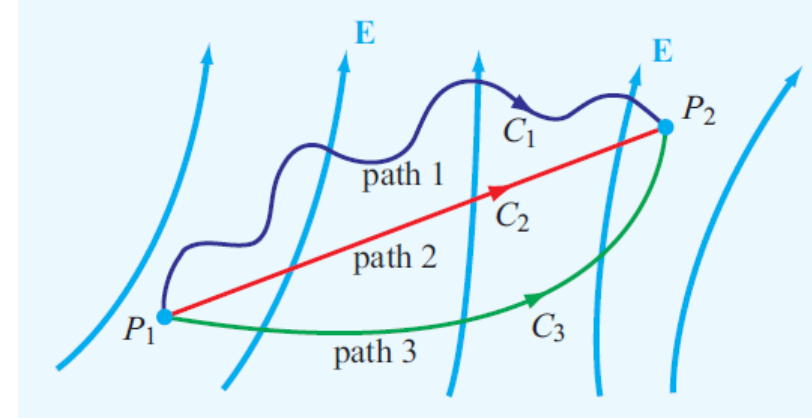
$$dV = \frac{dW}{q} = -\vec{E} \cdot d\vec{l} \left[\frac{J}{C} \text{ or } V \right].$$

- The potential difference between points A and B, corresponding to moving a point charge from point A to point B is obtained by integration along any path between the points:

$$V_{AB} = - \int_A^B \vec{E} \cdot d\vec{l}$$

Electric Potential (2/2)

- Independent of path; this follows immediately from the law of conservation of Energy.
- Potential is higher near positive charges and lower near negative charges.
- It's the “energy landscape” of an electric field.
- If V_{AB} is negative, there is a loss in potential energy in moving Q from A to B ; this implies that the work is being done by the field.
- If V_{AB} is positive, there is a gain in potential energy in the movement; an external agent performs the work.
- The term “**voltage**” is short for “voltage potential” and synonymous with electric potential.



- The line integral of the **electrostatic** field E around any closed contour C is zero:

$$V_{AA} = - \oint \vec{E} \cdot d\vec{l} = 0$$

- Recall Kirchhoff's Voltage Law
- Applying Stokes's theorem:

$$\oint \vec{E} \cdot d\vec{l} = \int \nabla \times \vec{E} \cdot d\vec{S} = 0$$

$$\nabla \times \vec{E} = 0$$

- One of the components of **Maxwell's Equations** for Electrostatic Conditions.
- Electrostatic fields are **conservative**.

Electric Potential – Point Charge

- If the electric field is caused by a point charge Q located at the origin, where

$$\vec{E} = \frac{Q}{4\pi\epsilon_0 r^2} \vec{a}_r$$

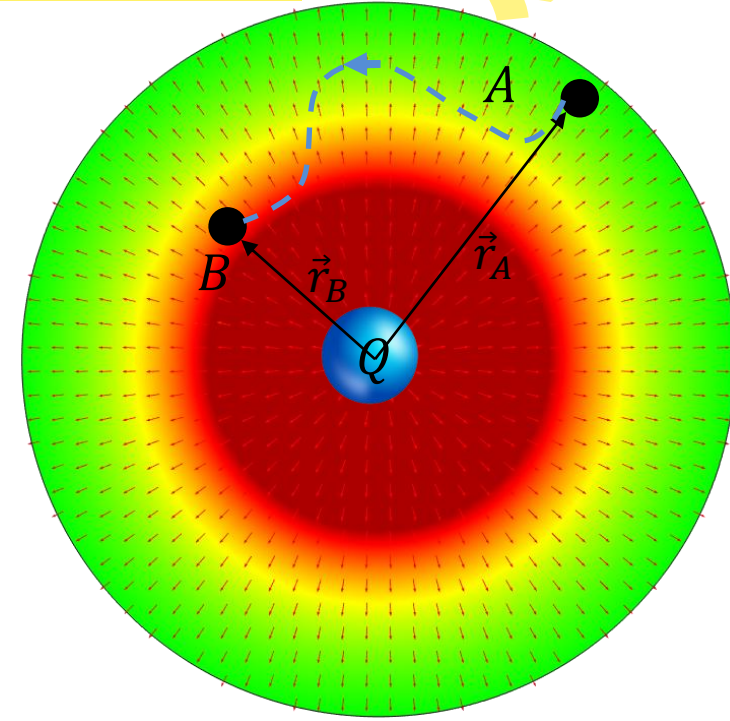
- The potential difference between points A and B located at $\vec{r}_A$ and $\vec{r}_B$ is:

$$V_{AB} = - \int_A^B \vec{E} \cdot d\vec{l} = - \int_A^B \left(\frac{Q}{4\pi\epsilon_0 r^2} \vec{a}_r \right) \cdot d\vec{l} = \frac{-Q}{4\pi\epsilon_0} \int_A^B \frac{1}{r^2} \vec{a}_r \cdot d\vec{l}$$

- But $d\vec{l} = dr\vec{a}_r + r d\theta \vec{a}_\theta + r \sin \theta d\phi \vec{a}_\phi$

$$V_{AB} = \frac{-Q}{4\pi\epsilon_0} \int_A^B \frac{1}{r^2} \vec{a}_r \cdot (dr\vec{a}_r + r d\theta \vec{a}_\theta + r \sin \theta d\phi \vec{a}_\phi) = \frac{-Q}{4\pi\epsilon_0} \int_A^B \frac{dr}{r^2} = \frac{-Q}{4\pi\epsilon_0} \left(\frac{-1}{r} \right)_{r_A}^{r_B}$$

$$V_{AB} = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{r_B} - \frac{1}{r_A} \right)$$



Electric Potential – Absolute Potential

- **Absolute Electric Potential** or Electric potential V at a point P at position $\vec{r}$ in space: Work done to move a unit charge from infinity to point P .

- In the previous equation set $A \rightarrow \infty, B \rightarrow P$
- Hence the absolute potential $V(\vec{r})$ at point $\vec{r}$ is:

$$V(\vec{r}) = - \int_{\infty}^P \vec{E} \cdot d\vec{l} = \frac{Q}{4\pi\epsilon_0} \frac{1}{r}$$

- If the charge Q is located at point $\vec{r}'$ instead of the origin:

$$V(\vec{r}) = \frac{Q}{4\pi\epsilon_0} \frac{1}{|\vec{r} - \vec{r}'|}$$

- If there are multiple point charges:

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \sum_i \frac{Q_i}{|\vec{r} - \vec{r}'_i|}$$

- For continuous charge distribution:

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_{l'} \frac{\rho_l}{|\vec{r} - \vec{r}'_l|} dl' \text{ line charge distribution}$$

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_{S'} \frac{\rho_s}{|\vec{r} - \vec{r}'_s|} dS' \text{ surface charge distribution}$$

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_{v'} \frac{\rho_v}{|\vec{r} - \vec{r}'_v|} dv' \text{ volume charge distribution}$$

Relationship between $\vec{E}$ and V

- Recall:

$$V = - \int \vec{E} \cdot d\vec{l}$$

which implies

$$dV = -\vec{E} \cdot d\vec{l}$$

but dV can be written as $dV = \nabla V \cdot d\vec{l}$ hence, we can establish:

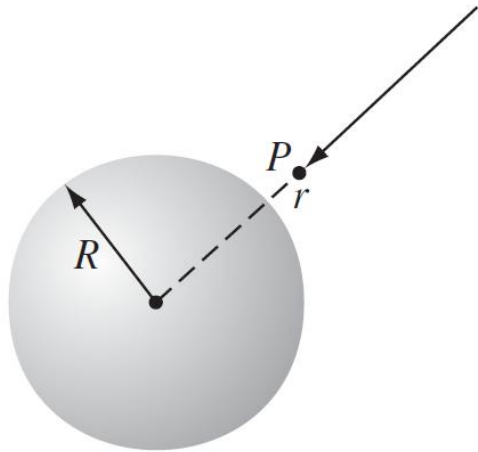
$$\vec{E} = -\nabla V$$

- The negative sign shows that the direction of $\vec{E}$ is opposite to the direction in which V increases; $\vec{E}$ is directed from higher to lower levels of V .



Electric Potential Examples

- Find the potential inside and outside a spherical shell of radius R that carries a uniform surface charge Q .



Solution

- From Gauss's law, the field inside of the sphere is zero while the field outside is:

$$\vec{E} = \frac{Q}{4\pi\epsilon_0 r^2} \vec{a}_r$$

- The potential for points outside the sphere $r > R$:

$$V(\vec{r}) = - \int_{\infty}^r \vec{E} \cdot d\vec{l}' = \frac{-Q}{4\pi\epsilon_0} \int_{\infty}^r \frac{1}{r'^2} \vec{a}_r \cdot dr' \vec{a}_r = \frac{-Q}{4\pi\epsilon_0} \int_{\infty}^r \frac{1}{r'^2} dr' = \frac{-Q}{4\pi\epsilon_0} \left[-\frac{1}{r'} \right]_{\infty}^r = \frac{Q}{4\pi\epsilon_0 r}$$

- To find the potential inside the sphere ($r < R$), we break the integral into two pieces:

$$V(\vec{r}) = - \int_{\infty}^R \vec{E} \cdot d\vec{l}' - \int_R^r \vec{E} \cdot d\vec{l}' = \frac{Q}{4\pi\epsilon_0 R} - \int_R^r 0 \cdot d\vec{l}' = \frac{Q}{4\pi\epsilon_0 R}$$

Observations

- V is not zero inside the shell, even though the field vanishes! V is a constant in this region, which is confirmed by $-\nabla V = \vec{E} = 0$.
- In problems of this type, you must always work your way in from the reference point; that's where the potential is "nailed down."
- It is tempting to suppose that you could figure out the potential inside the sphere on the basis of the field there alone, but this is false: the potential inside the sphere is sensitive to what's going on outside the sphere as well.
- If a second charged shell is placed out at radius $r > R$, the potential inside R would change, even though the field would still be zero.
- Gauss's law guarantees that charge exterior to a given point (that is, at larger r) produces no net field at that point, provided it is spherically or cylindrically symmetric, but there is no such rule for potential, when infinity is used as the reference point.

Electric Dipole

- An **electric dipole** is formed when two point charges of equal magnitude but opposite sign are separated by a small distance.
- Appears in common in molecules (like water) and in antenna design.
- Acts like a tiny electric bar magnet with north and south poles.
- Consider a dipole consisting of two equal and opposite charges $+Q$ and $-Q$ separated by a small distance, d .
- We define the **Dipole Moment**: $\vec{p} = Q\vec{d}$, a vector directed from negative $-Q$ to positive charge $+Q$.
- The potential due to the dipole is:

$$V = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{r_1} - \frac{1}{r_2} \right) = \frac{Q}{4\pi\epsilon_0} \left(\frac{r_2 - r_1}{r_1 r_2} \right)$$

- If $r \gg d$, $r_2 - r_1 \approx d \cos \theta$, $r_1 r_2 \approx r^2$, therefore the potential becomes:

$$V = \frac{Q}{4\pi\epsilon_0} \frac{d \cos \theta}{r^2}$$

- Since $d \cos \theta = \vec{d} \cdot \vec{a}_r$, where $\vec{d} = d\vec{a}_z$, if we define $\vec{p} = Q\vec{d}$ as the **Dipole Moment**, the potential becomes:

$$V = \frac{Q\vec{d} \cdot \vec{a}_r}{4\pi\epsilon_0 r^2} = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \vec{a}_r}{r^2}$$

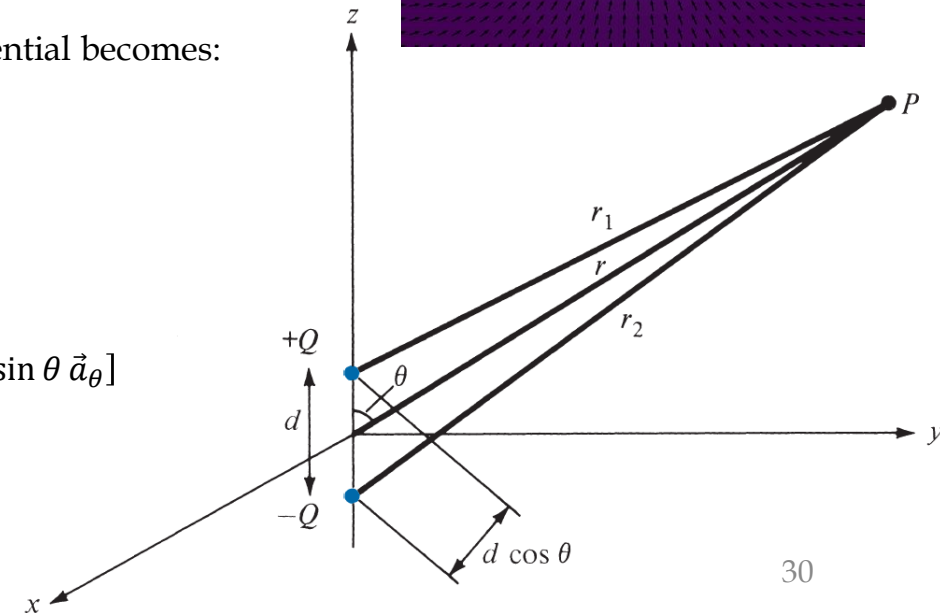
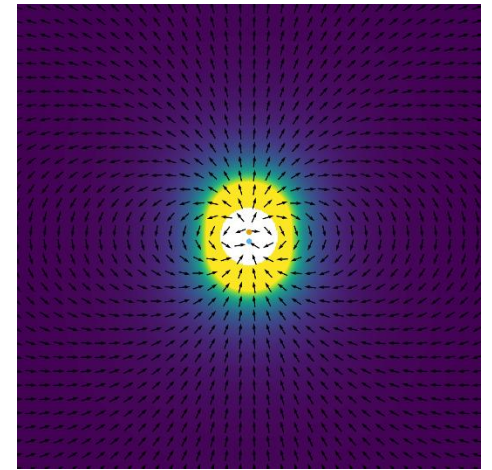
- If the dipole center is not at the origin but at r' ,

$$V = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3}$$

- The electric field due to the dipole with center at the origin, can be obtained as:

$$\begin{aligned} \vec{E} = -\nabla V &= - \left[\frac{\partial V}{\partial r} \vec{a}_r + \frac{1}{r} \frac{\partial V}{\partial \theta} \vec{a}_\theta \right] = \frac{Qd \cos \theta}{2\pi\epsilon_0 r^3} \vec{a}_r + \frac{Qd \sin \theta}{4\pi\epsilon_0 r^3} \vec{a}_\theta = \frac{Qd}{4\pi\epsilon_0 r^3} [2 \cos \theta \vec{a}_r + \sin \theta \vec{a}_\theta] \\ &= \frac{p}{4\pi\epsilon_0 r^3} [2 \cos \theta \vec{a}_r + \sin \theta \vec{a}_\theta] \end{aligned}$$

- The field intensity drops as $1/r^3$, faster than that of a single charge ($1/r^2$).



Electric Dipole - Exercises

1. Two dipoles $-5\vec{a}_z \text{ nCm}$ and $9\vec{a}_z \text{ nCm}$ are located at $(0,0,-2)$ and $(0,0,3)$ respectively. Find the potential at the origin.
2. An electric dipole of $100\vec{a}_z \text{ pCm}$ is located at the origin. Find V and $\vec{E}$ at points $(0,0,10)$ and $\left(1, \frac{\pi}{3}, \frac{\pi}{2}\right)$.



Energy in Electrostatic Fields (1/2)

- Building an electric field requires **work** – hence energy is stored within it.
- Energy can be visualized as the “**tension**” between electric charges.
- In devices like capacitors, this stored energy can later be released as useful electrical power.
- To determine the energy present in an assembly of charges, let us first determine the amount of work necessary to assemble them.
- Suppose we wish to position three point charges Q_1 , Q_2 , and Q_3 in an initially empty space shown shaded.
- No work is required to transfer Q_1 from infinity to P_1 because the space is initially charge free and there is no electric field: $W_1 = 0$.
- The work done in transferring Q_2 from infinity to P_2 is equal to the product of Q_2 and the potential V_{21} at P_2 due to Q_1 : $W_2 = Q_2 V_{21}$.
- Similarly, the work done in positioning Q_3 at P_3 is equal to $Q_3(V_{32} + V_{31})$, where V_{32} and V_{31} are the potentials at P_3 due to Q_2 and Q_1 , respectively: $W_3 = Q_3(V_{32} + V_{31})$
- Hence the total work done in positioning the three charges is:

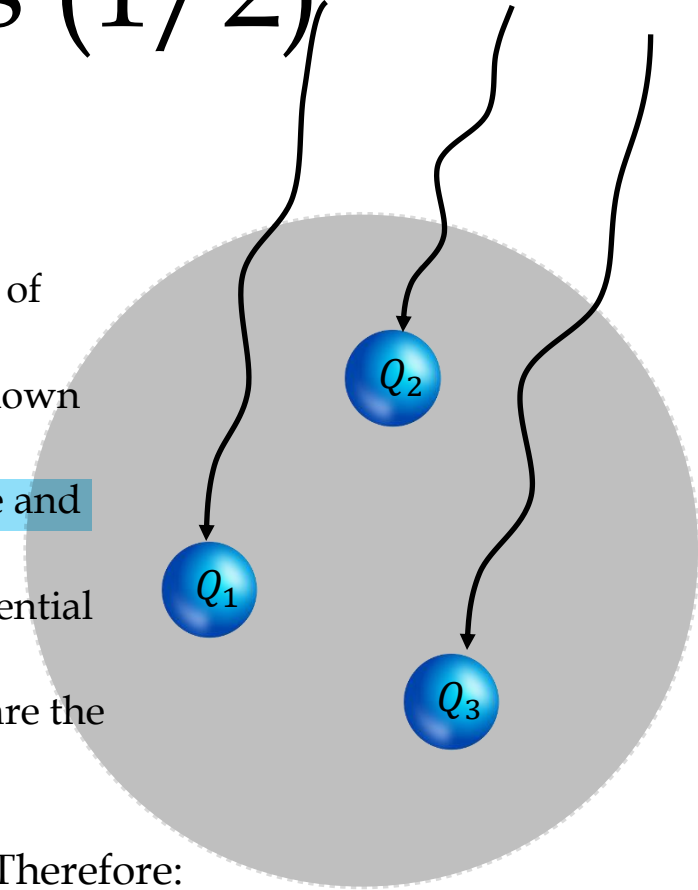
$$W_E = W_1 + W_2 + W_3 = 0 + Q_2 V_{21} + Q_3(V_{31} + V_{32})$$

- If the charges were positioned in reverse order:

$$W_E = W_3 + W_2 + W_1 = 0 + Q_2 V_{23} + Q_1(V_{12} + V_{13})$$

- Adding the two equations:

$$2W_E = Q_1(V_{12} + V_{13}) + Q_2(V_{21} + V_{23}) + Q_3(V_{31} + V_{32}) = Q_1 V_1 + Q_2 V_2 + Q_3 V_3$$



- Therefore:

$$W_E = \frac{1}{2} (Q_1 V_1 + Q_2 V_2 + Q_3 V_3)$$

- where V_1 , V_2 , and V_3 are total potentials at P_1 , P_2 , and P_3 , respectively.

Energy in Electrostatic Fields (2/2)

- In general, if there are n point charges, we have:

$$W_E = \frac{1}{2} \sum_{k=1}^n Q_k V_k$$

- If, instead of point charges, the region has a continuous charge distribution

$$W_E = \frac{1}{2} \int_{l'} \rho_L V dl' \text{ for line charge}$$

$$W_E = \frac{1}{2} \int_{S'} \rho_S V dS' \text{ for surface charge}$$

$$W_E = \frac{1}{2} \int_{v'} \rho_v V dv' \text{ for volume charge}$$

- Since $\rho_v = \nabla \cdot \vec{D}$ and , the Energy stored can be written as:

$$W_E = \frac{1}{2} \int_{v'} (\nabla \cdot \vec{D}) V dv'$$

- But for any vector $\vec{A}$ and scalar V , we have the identity $\nabla \cdot V\vec{A} = \vec{A} \cdot \nabla V + V(\nabla \cdot \vec{A})$ or $(\nabla \cdot \vec{A})V = \nabla \cdot V\vec{A} - \vec{A} \cdot \nabla V$. Applying the identity we get

$$W_E = \frac{1}{2} \int_{v'} (\nabla \cdot V\vec{D}) dv' - \frac{1}{2} \int_{v'} (\vec{D} \cdot \nabla V) dv'$$

- By applying divergence theorem to the first term on the right-hand side of this equation, we have

$$W_E = \frac{1}{2} \oint_{S'} (V\vec{D}) \cdot d\vec{S}' - \frac{1}{2} \int_{v'} (\vec{D} \cdot \nabla V) dv'$$

- Recall that V varies as $\frac{1}{r}$ and $\vec{D}$ as $\frac{1}{r^2}$ for point charges; V varies as $\frac{1}{r^2}$ and $\vec{D}$ as $\frac{1}{r^3}$ for dipoles; and so on. Hence, $V\vec{D}$ in the first term on the right-hand side must vary at least as $\frac{1}{r^3}$ while $d\vec{S}'$ varies as $\frac{1}{r^2}$. Consequently, the first integral must tend to zero as the surface S becomes large. Hence, the equation reduces to:

$$W_E = -\frac{1}{2} \int_{v'} (\vec{D} \cdot \nabla V) dv' = \frac{1}{2} \int_{v'} (\vec{D} \cdot \vec{E}) dv' = \frac{1}{2} \int_{v'} (\epsilon_0 \vec{E} \cdot \vec{E}) dv'$$

$$W_E = \frac{1}{2} \int_{v'} \epsilon_0 |\vec{E}|^2 dv'$$

- From this, we can define electrostatic energy density w_E (in $\frac{J}{m^3}$) as:

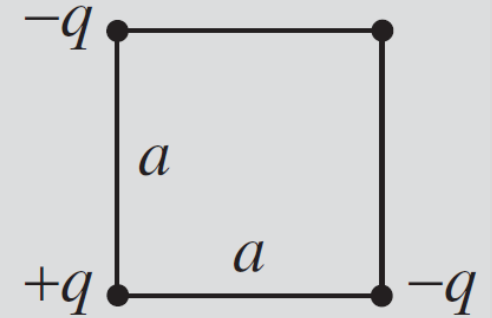
$$w_E = \frac{dW_E}{dv} = \frac{1}{2} \vec{D} \cdot \vec{E} = \frac{1}{2} \epsilon_0 |\vec{E}|^2 = \frac{|\vec{D}|^2}{2\epsilon_0}$$



Energy in Electrostatic Fields - Exercises

1. Point Charges

- Three charges are situated at the corners of a square (side length a), as shown in the figure. How much work does it take to bring in another charge, $+q$, from far away and place it in the fourth corner?
- How much work does it take to assemble the entire configuration of four charges?



- Assuming the plates as infinite planes, derive formulation for energy stored inside a parallel plate capacitor with voltage V across the plates and that has stored charge Q .
- Find the energy stored in a uniformly charged solid sphere of radius R and charge q . Do it in two different ways:
 - Using the potential and charge distribution.
 - Using the Electric Field
- Consider two concentric spherical shells, of radii a and b . Suppose the inner one carries a charge q , and the outer one a charge $-q$ (both of them uniformly distributed over the surface). Calculate the energy of this configuration in two different ways, similar to problem 3 above.

Energy in ... - Exercises (Solution 1/2)

1.a) Assuming the Q_1 is at the origin, the work done to bring unit charge from infinity to point P $\vec{r} = a\vec{a}_x + a\vec{a}_y$ is the absolute potential $V(\vec{r})$ at the point:

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left(\frac{Q_1}{R_1} + \frac{Q_2}{R_2} + \frac{Q_3}{R_3} \right)$$

where: $\vec{r}'_1 = 0$, $\vec{r}'_2 = a\vec{a}_y$, $\vec{r}'_3 = a\vec{a}_x$ and $R_1 = |\vec{r} - \vec{r}'_1| = |a\vec{a}_x + a\vec{a}_y - 0| = \sqrt{2}a$, $R_2 = a$, $R_3 = a$

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{\sqrt{2}a} + \frac{-q}{a} + \frac{-q}{a} \right) = \left(\frac{\sqrt{2} - 4}{8} \right) \frac{q}{\pi\epsilon_0 a}$$

The work done to bring charge q from infinity to point P is therefore:

$$W_E = qV(\vec{r}) = \left(\frac{\sqrt{2} - 4}{8} \right) \frac{q^2}{\pi\epsilon_0 a}$$

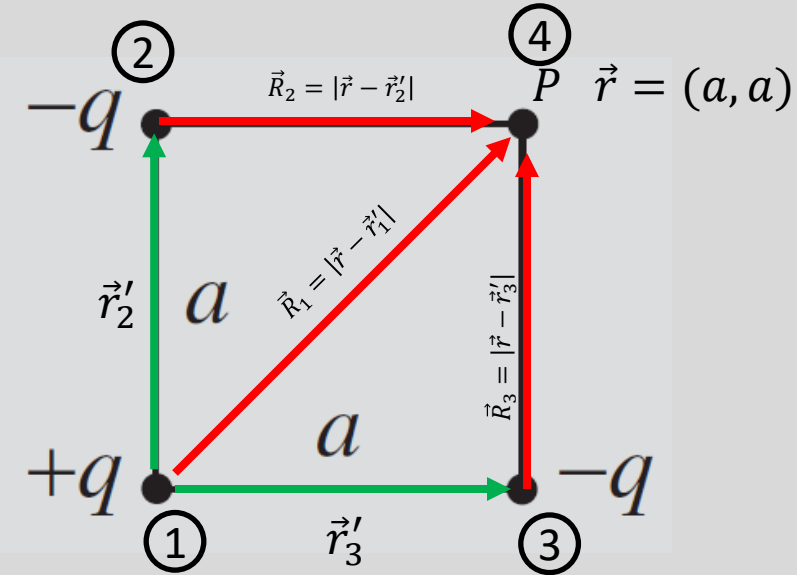
The result is negative (the field does the work).

- The total work done to assemble all four charges is:

$$W_E = \frac{1}{2} (Q_1 V_1 + Q_2 V_2 + Q_3 V_3 + Q_4 V_4)$$

where:

$$\begin{aligned} V_1 &= \frac{1}{4\pi\epsilon_0} \left(\frac{Q_2}{R_{12}} + \frac{Q_3}{R_{13}} + \frac{Q_4}{R_{14}} \right) = \frac{1}{4\pi\epsilon_0} \left(\frac{-q}{a} + \frac{-q}{a} + \frac{q}{\sqrt{2}a} \right) = \frac{q}{4\pi\epsilon_0 a} \left(-2 + \frac{1}{\sqrt{2}} \right) \\ V_2 &= \frac{1}{4\pi\epsilon_0} \left(\frac{Q_1}{R_{21}} + \frac{Q_3}{R_{23}} + \frac{Q_4}{R_{24}} \right) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{a} + \frac{-q}{\sqrt{2}a} + \frac{q}{a} \right) = \frac{q}{4\pi\epsilon_0 a} \left(2 - \frac{1}{\sqrt{2}} \right) \\ V_3 &= \frac{1}{4\pi\epsilon_0} \left(\frac{Q_1}{R_{31}} + \frac{Q_2}{R_{32}} + \frac{Q_4}{R_{34}} \right) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{a} + \frac{-q}{\sqrt{2}a} + \frac{q}{a} \right) = \frac{q}{4\pi\epsilon_0 a} \left(2 - \frac{1}{\sqrt{2}} \right) \\ V_4 &= \frac{1}{4\pi\epsilon_0} \left(\frac{Q_1}{R_{41}} + \frac{Q_2}{R_{42}} + \frac{Q_3}{R_{43}} \right) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{\sqrt{2}a} + \frac{-q}{a} + \frac{-q}{a} \right) = \frac{q}{4\pi\epsilon_0 a} \left(-2 + \frac{1}{\sqrt{2}} \right) \end{aligned}$$



therefore:

$$\begin{aligned} W_E &= \frac{1}{2} (Q_1 V_1 + Q_2 V_2 + Q_3 V_3 + Q_4 V_4) \\ &= \frac{1}{2} \left(\frac{q^2}{4\pi\epsilon_0 a} \left(-2 + \frac{1}{\sqrt{2}} \right) - \frac{q^2}{4\pi\epsilon_0 a} \left(2 - \frac{1}{\sqrt{2}} \right) - \frac{q^2}{4\pi\epsilon_0 a} \left(2 - \frac{1}{\sqrt{2}} \right) + \frac{q^2}{4\pi\epsilon_0 a} \left(-2 + \frac{1}{\sqrt{2}} \right) \right) \\ &= \frac{q^2}{8\pi\epsilon_0 a} \left(\left(-2 + \frac{1}{\sqrt{2}} \right) - \left(2 - \frac{1}{\sqrt{2}} \right) - \left(2 - \frac{1}{\sqrt{2}} \right) + \left(-2 + \frac{1}{\sqrt{2}} \right) \right) \\ &= \frac{(-4 + \sqrt{2})q^2}{4\pi\epsilon_0 a} \end{aligned}$$

- negative $\rightarrow$ net energy released while assembling

Energy in ... - Exercises (Solution 2/2)

The electric field between the plates of the capacitor, assuming infinite planes:

$$E = \frac{\rho_s}{\epsilon_0}$$

The energy stored between the plates is then:

$$W_E = \frac{1}{2} \int_{v'} \epsilon_0 |\vec{E}|^2 dv' = \frac{1}{2} \epsilon_0 \frac{\rho_s^2}{\epsilon_0^2} Ad = \frac{\rho_s^2 A^2 d}{2\epsilon_0 A} = \frac{Q^2 d}{2\epsilon_0 A}$$

Since the electric field is constant, the voltage across the terminals is:

$$V = - \int \vec{E} \cdot d\vec{l} = Ed = \frac{\rho_s d}{\epsilon_0} = \frac{\rho_s Ad}{\epsilon_0 A} = \frac{Qd}{\epsilon_0 A}$$

Therefore:

$$W_E = \frac{Q^2 d}{2\epsilon_0 A} = \frac{Q}{2} \frac{Qd}{\epsilon_0 A}$$
$$W_E = \frac{1}{2} QV$$



Summary and Concept Map

- Key ideas in electrostatics:
 - Coulomb's Law → explains force between charges.
 - Electric Field → shows influence in space.
 - Gauss's Law → simplifies symmetric problems.
 - Potential → relates to energy and work.
 - Dipoles → foundation for polarization and antenna theory.
 - Energy → quantifies stored power in fields.
- Together, these concepts form the static foundation of Maxwell's Equations.



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